

HELENA FU

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EDUCATION

Tufts University

Medford, MA

B.S. Computer Science, B.A. Mathematics, Minor: AI Development

May 2027

GPA: 3.93 / 4.00

Honors: 6x Tufts Dean's List, AI Gardner Memorial Endowment Scholarship (Society of Women Engineers)

Relevant Coursework: Machine Learning, Database Systems, Operating Systems, Algorithms, Data Structures, Intro to Security

EXPERIENCE

Tesla

Austin, TX

Software Engineering Intern

June 2026 – Dec 2026

- Architected and shipped real-time voice-to-voice AI with React, Wails, Go, Kubernetes, and MongoDB, scaling to 30K users.
- Built MCP agents (Jira, Teams, etc.) using Go, python, gRPC/protobuf, handling 3M tool calls/week to automate workflows.
- Used AI tools (Claude CLI, Cursor) to accelerate prototyping, code review, and debugging.

Mad Realities

New York City, NY

Software Engineering Intern

June 2025 – Aug 2025

- Developed SMS-based LLM using Next.js, Cloudflare, and Docker, engaging 3K+ users and boosting interaction by 40%.
- Improved information retrieval accuracy by 85% via embedding optimization, prompt engineering, and RAG fine-tuning.
- Shipped data analysis pipeline and dashboard using TypeScript, SQL, and Palantir Foundry, cutting report time by 90%.

Acadia Analytics Inc

Medford, MA

Quantitative Developer

Jan 2025 – May 2025

- Architected trading service in Python/Flask, serving real-time predictions to 50+ users with 99.5% uptime over 4 months.
- Built a pytest suite integrated into the deployment pipeline, achieving 85% code coverage across unit, integration, and end-to-end tests and preventing regressions before production releases.

Tufts Human-Robot Interaction Lab

Medford, MA

Research Assistant

Feb 2024 – May 2025

- Extended robot architecture with a Java + Redis caching module, reducing natural language processing latency by 46%.
- Built a real-time preprocessing service in Python, improving cognitive model input quality by 30%.

PROJECTS

MIT Lincoln Lab Capstone – Autonomous Vehicle Awareness | PyTorch, Python, MATLAB, Linux

- Led a 5-person team to build a real-time radar-camera fusion system on embedded hardware, fusing CNN vision with FMCW radar detection to hit < 800ms end-to-end latency.
- Built live dashboard (Next.js + Flask) streaming synchronized multi-sensor data, fused detections, and profiling metrics.

DSCVR.AI Research App | Next.js, Docker, AWS Bedrock

- Created AI writing assistant that cuts down research and drafting time by using RAG to surface personalized, contextually relevant suggestions from user's knowledge base in real time.
- Earned Cornell NYC AI **Hackathon Winner** and "Most App Store Ready" Award among 30+ competing teams.

PUBLICATIONS

Aygun, A., **Fu, H.**, Krause, E., Scheutz, M. "Online Human Workload Detection for Behavior Adaptation in Autonomous Robot Teammates" — *International Conference on Social Robotics 2026*, **Best Paper Finalist**.

TECHNICAL SKILLS

Languages: Python, Go, Bash, JavaScript, TypeScript, Java, C++, SQL, HTML/CSS

Infrastructure & DevOps: Kubernetes, Git/GitHub, Docker, AWS, Linux, Google Cloud Platform, CI/CD, REST APIs

Full stack: Flask, Node.js, Express, React, Next.js, Tailwind CSS, Wails

Databases & Data: SQL, MongoDB, Redis, Pinecone, Palantir Foundry, Tableau